FIELD PROJECT REPORT-II

on

PASSPORT VERIFICATION USING RFID CARD

Submitted by

SK.BAJI	**201FA05057**
SK.ARSHIYA	**201FA05075**
G.DEVI SAI SUNANDA	**211LA05026**
A.RAKESH	**211LA05032**



Under the esteemed guidance of

Dr. K. Venkata Kishore (Associate Professor)

Dr. M.Pachiyannan (Associate Professor)

Ms. Naga Jyothi Sree

Department of ECE



(ACCREDITED BY NAAC WITH '**A**+' GRADE**)**

Vadlamudi,Guntur,Andhra Pradesh,India-522213

2021-2022

CERTIFICATE

This is to certify that the field project entitled "PASSPORT VERIFICATION USING RFID CARDS" that is being submitted by BAJI, ARSHIYA, G.DEVI SAI SUNANDA, RAKESH bearing Regd. Nos. 201FA05057, 201FA05075, 211LA05026, 211LA05032 in partial fulfillment for the award of II-year, II-semester B. Tech degree in Electronics and Communication Engineering to Vignan's Foundation for Science, Technology and Research, is a record of work carried outby him/her under the guidance of ECE Department.

Signature of the faculty
Dr. K. Venkata Kishore
Associate Professor

Signature of Head of the Department
Dr. T. Pichaiah, M.E., Ph. D, MIEEE
Professor, HOD, ECE

Dr.M.Pachiyannan
Associate Professor

TABLE OF CONTENTS

1. Abstract
2. Introduction
3. Objectives
4. Literature review
5. Theoretical analysis of proposed work
6. Proposed design
7. Implementation of proposed work
8. Components
9. Working
10. Results and analysis
11. Conclusion
12. Future scope
13. Acknowledgement
14. References

1. ABSTRACT

In this digital world, RFID technology is applied to many applications in different fields such as transportation, health-care, industries etc. This technology along with Internet of things facilitates wireless identification using active and passive tags with suitable readers. RFID technology is applied for passport verification system to authenticate the passport holder. This avoids forgery and manual work associated with traditional passport verification system. The passport checker checks the passenger's passport by means of e-passport.

The Electronic passport is the digital version of the paper passport to provide stronger identity authentication. Passport verification and checking which a very time-consuming process. To be ease identity checks, lessen the amount of human errors, protect against manipulation of travel documents and improve border security issues hence, new passport turned out to be much more intrusive than the traditional one.

The proposed system simplifies this process with RFID card where the unique identification number is stored which corresponds to the information of the person. The information includes the name, nationality, address etc. along with attach the copy of the required certificates required according to the application are managed or collected in paper. The information is transferred to computer with the help of RF transceiver. This proposed system uses Radio Frequency Identification (RFID) is a technology that uses wireless communication for identification purposes. The key characteristic that differentiates one RFID application from another is the purpose of identification.

2. INTRODUCTION

The passport verification system using RFID, as it is sometimes called, represents a bold initiative in the deployment of two new technologies: Radio Frequency Identification (RFID) and biometrics. System are wireless technology for automatic identification. They bring forth the era of next generation ID cards. Several national governments plan to deploy identity cards integrating RFID and biometrics for domestic use. We explore the privacy and security and other issues of the e-passport in this article cards.

This passport verification system using RFID technology, promise to reduce fraud, ease identity checks, enhance security. This project will automate the whole process of passport verification by using RFID technology. Passports will be issued using RFID tags or cards that electronically contain all important data that must be included in a passport. The passport holder will swipe or scan their RFID tags or cards into the RFID reader that will compare the scanned data to the data of the passport holder in the system.

The RFID-based passport verification will strengthen the security and privacy of the passport holders and avoid all serious threats. The drawback of paper passports are that they do not have privacy and anyone can physically access or duplicate the passport. When the RFID tag with passport holder is brought near the RFID reader, the data is read and displayed on the LCD. If the data matches with the data in the program memory, it displays a valid message as defined.

3. OBJECTIVES

The objective is for eliminating the conventional paper passport booklet using RFID cards and to improve the security by creating a stronger link between the passport and its holder.

Primary Objective

To develop an e-passport system using RFID cards.

Secondary Objectives

To make it almost impossible to alter a document for use in any other ways.

4. LITERATURE REVIEW

1. An E-Passport holder holds an electronic chip such as RFID. This device has several advantages such as simple operation, easy to carry up, less cost, wireless control and so on.
2. The chip holds the similar data that is printed on the passport information page such as the passport holder's name, date of birth, and other biographic information.

5. THEORETICAL ANALYSIS OF PROPOSED WORK

1. To overcome the above-mentioned problems, the researchers of the study proposed an RFID-enabled passport verification.
2. The main functionality of this project is to access and automate the whole process of the passport details of a passport holder through RFID technology.
3. Passports will be issued using RFID tags or cards that electronically contain all important data that must be included in a passport.
4. The passport holder will swipe or scan their RFID tags or cards into the RFID reader that will compare the scanned data to the data of the passport holder in the system.
5. The RFID-based passport verification will strengthen the security and privacy of the passport holders and avoid all serious threats.

6. PROPOSED DESIGN

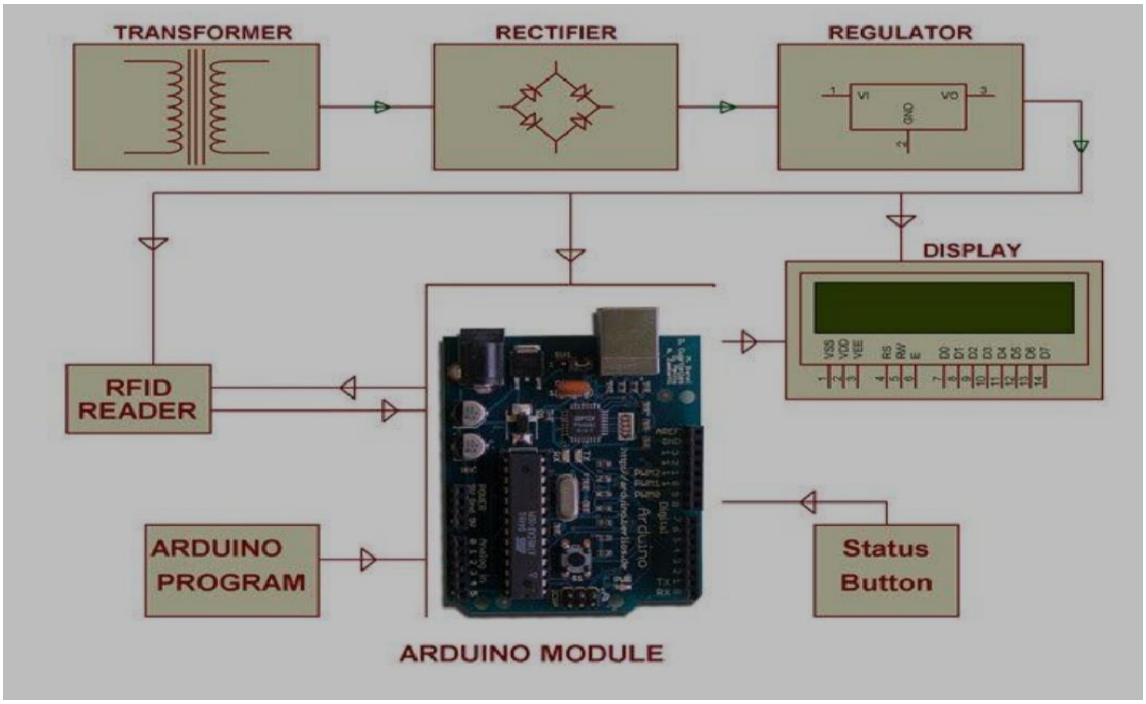


Figure 1

7. IMPLEMENTATION OF PROPOSED WORK

1. Passport holders will have RFID tags and airports will have an RFID scanner.
2. The project is limited to storing the passport details in RFID tags and details will be compared to the data stored in the computer after scanning.

8. HARDWARE COMPONENTS

1. ARDUINO UNO

Arduino UNO is a microcontroller board based on the **ATmega328P**. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.



Fig 2: ARDUINO UNO

2. **RFID**

RFID (radio frequency identification) is a form of wireless communication that incorporates the use of electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to uniquely identify an object. The RFID system comprises the RFID tag/card, RFID reader, backend database and a control unit. RFID system consists of three components: a scanning antenna, transceiver and a transponder. The RFID reader communicates with the RFID tag through tag interrogation.



Fig 3: RFID READER

3. **LCD:**

It stands for Liquid Crystal Display. A LCD is a flat-panel electronic visual display that makes use of the light modulating properties of liquid crystals.These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc**.**

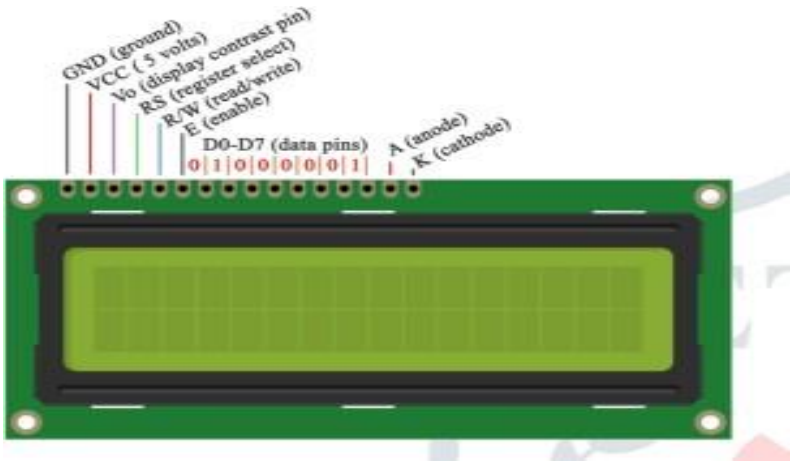


Fig 4: LCD DISPLAY

9. WORKING

1. Basically the RFID structure comprises of three elements which are: An antenna or coil and Transponder (RF Tag) electronically programmed with unique information.
2. Transceiver (with decoder) these elements communicate using radio signals which carry data either unidirectional or bidirectional.
3. When a transponder gets into a red zone, its contents are captured by the reader and can then be transferred through standard interfaces to host devices such as a computer (Arduino).
4. Arduino is coded in such a way that it displays in LCD whether to allows or deny the access of the person.

10. RESULTS

- Passport checker checks the passenger's passport by means of E passport embedded with RFID tag.
- Reduces the time taken to receive the passport after verification.

11. CONCLUSION

This project gives clear idea about the Electronic passport system which is much more beneficial for the airports and universities. It also reduces the burden of documentation as well as it reduces the time consumption. We analysed the major current and potential uses of RFID in identifying documents and the most important feature of this project is security, this will make the system centralized. The security of the system can be further increased by adding biometric information such as fingerprints, palm scan, iris scan, digital signature and another active authentication in the passport system

12. FUTURE SCOPE

1. One card Policy throughout country and also the security of the system can be further increased by adding biometric information such as fingerprints, palm scan, iris scan, digital signature and other active authentication in the passport system.
2. Face recognition and OTP option or pin can be provided can be used for better security purpose.
3. All data can be clouded/ stored in data base for verification.

13. ACKNOWLEDGEMENT

We thank the department of ECE, VFSTR for including Field Project in the curriculum. We also thank our faculty for guiding us in implementation of this project work.

14. REFERENCES

- G. Matthew Ezovski, Steve E. Watkins, ―The Electronic Passport and the Future of Government Issued RFID-Based Identification‖ 2007 IEEE International Conference on RFID Gaylord Texan Resort, Grapevine, TX, USA March 26-28, 2007.
- Marci Meingast, Jennifer King, and Deirdre K. Mulligan, "Security and Privacy Risks of Embedded RFID in Everyday Things: the e-Passport and Beyond," Journal of Communications, vol. 2, no. 7, pp. 36-48, 2007.
- K. Ouafi and R. C.-W. Phan, "Privacy of recent RFID authentication protocols," 4th International Conference on Information Security Practice and Experience – ISPEC 2008, ser. Lecture Notes in Computer Science, vol. 4991. Sydney, Australia: Springer, April 2008, pp. 263–277.
- Shivani Kundra, Aman Dureja, Riya Bhatnagar, "The study of recent technologies used in E-passport system", IEEE (GHTC-SAS), September 26-27, 2014.